Math 141, F11 Name: Answer

Quiz 7 Section: SSN: xxx-xx-

Instructions: There is totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1.(6 points) Find the derivatives of the following functions.

(a) $y = \log_2(1-3x)$

$$\frac{dy}{dx} = \frac{1}{(1-3x)\ln 2}\frac{d}{dx}[1-3x]$$
$$= \frac{-3}{(1-3x)\ln 2}$$

(b) $y = \ln(xe^{x^2})$

$$y = \ln x + \ln e^{x^2}$$
$$= \ln x + x^2 \ln e$$
$$= \ln x + x^2$$
$$\Rightarrow \frac{dy}{dx} = \frac{1}{x} + 2x$$

Problem 2.(4 points) Use *logarithmic differentiation* to find the derivative of the function $y = x^{\tan x}$.

step 1. $\ln y = \ln(x^{\tan x})$

$\ln y = \tan x \ln x$

step 2. $\frac{d}{dx}[\ln y] = \frac{d}{dx}[\tan x \ln x]$

$$\frac{1}{y}\frac{dy}{dx} = \tan x \cdot \left(\frac{1}{x}\right) + \ln x \cdot \sec^2 x$$
$$= \frac{\tan x}{x} + \sec^2 x \ln x$$

step 3. $\frac{dy}{dx} = y\left(\frac{\tan x}{x} + \sec^2 x \ln x\right)$

$$= x^{\tan x}\left(\frac{\tan x}{x} + \sec^2 x \ln x\right).$$